

Federated Learning and Naturalistic Driving

Lapland University of Applied Sciences

Dr. Manuel García Fernández

manuel.garcia2@universidadeuropea.es

Ve más allá

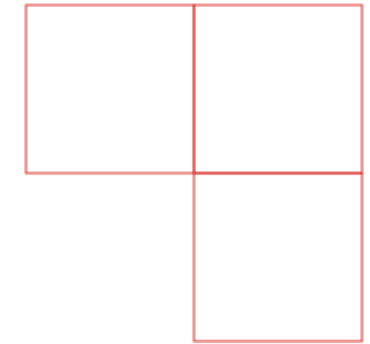
Index

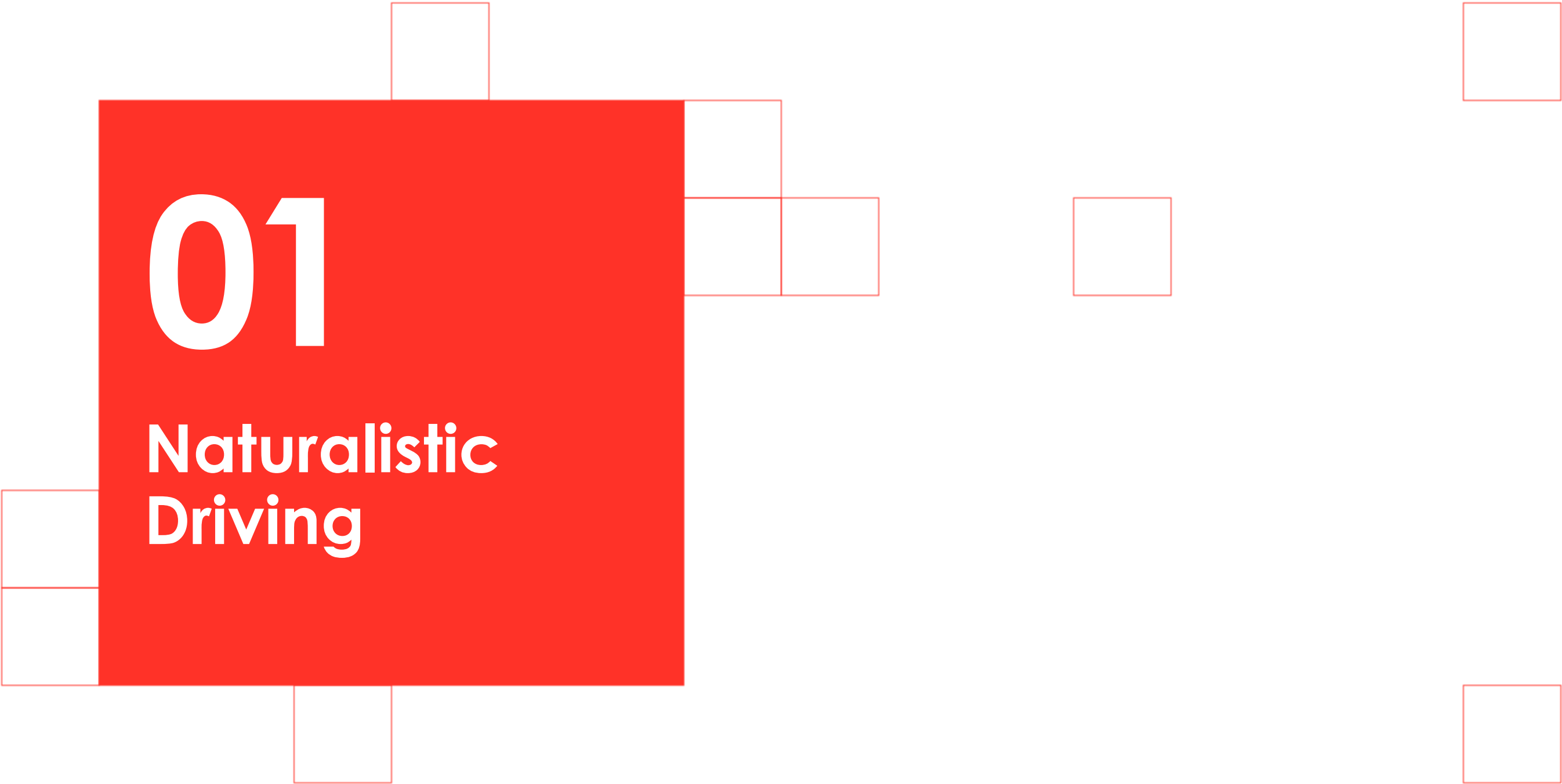
- Naturalistic Driving
- Data collection systems
- Federated Learning

01

02

03



A decorative graphic consisting of a large red square in the center-left. To its left are two stacked white squares with red borders. Above the red square is one white square with a red border. To its right are two stacked white squares with red borders, and further right is another single white square with a red border. Below the red square is one white square with a red border. On the far right of the slide is a single white square with a red border.

01

Naturalistic Driving



Naturalistic Driving

Autonomous driving can be divided into two approaches:

- Multistage: These types of approaches rely on dividing driving into 5 different stages:
 - Perception.
 - Localization.
 - Prediction.
 - Planning.
 - Control.
- Naturalistic (end-to-end): Single and unified Deep Neural Network that learns from real driving data.

Whereas Multistage Autonomous driving is easily interpretable and modular, it **requires a lot of fine-tuning and requires to handle explicitly rules for complex interactions.**

On the other side, naturalistic driving handles implicitly complex and diverse driving situations, but,

Naturalistic Driving

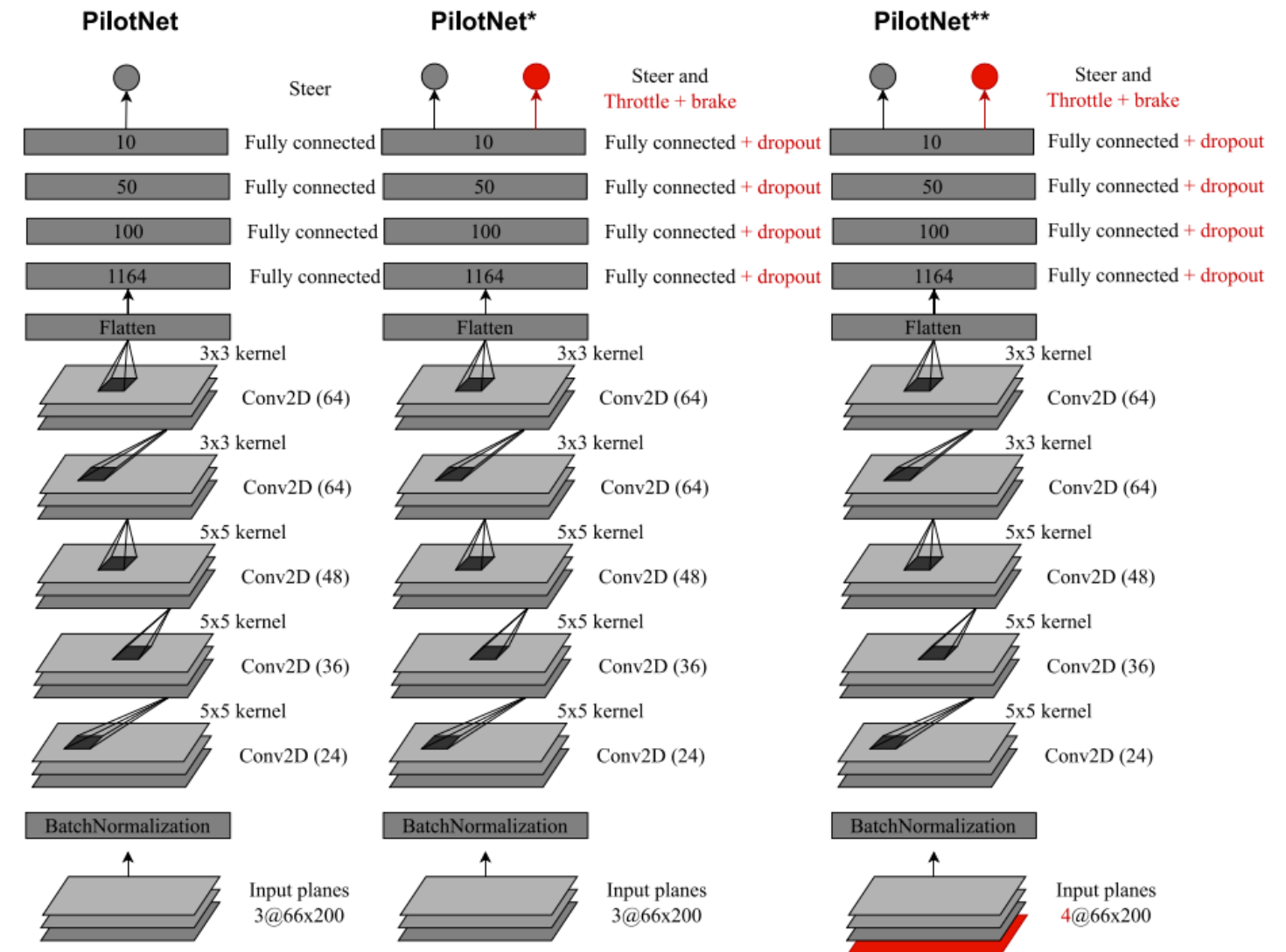
The simple Naturalistic Driving neural network is PilotNet. PilotNet, contains 5 Convolutional layers stacked with several linear layers.

PilotNet has some evolutions that can tackle also throttle and brake pedal actuation.

It aims to learn directly from the pixel map of an image the steer command of the car.

<https://arxiv.org/pdf/2010.08776>

<https://www.sciencedirect.com/science/article/pii/S0925231224006453>



Naturalistic Driving

Despite its simplicity, PilotNet can handle easily usual driving situations.

Nevertheless PilotNet lacks planning capabilities.

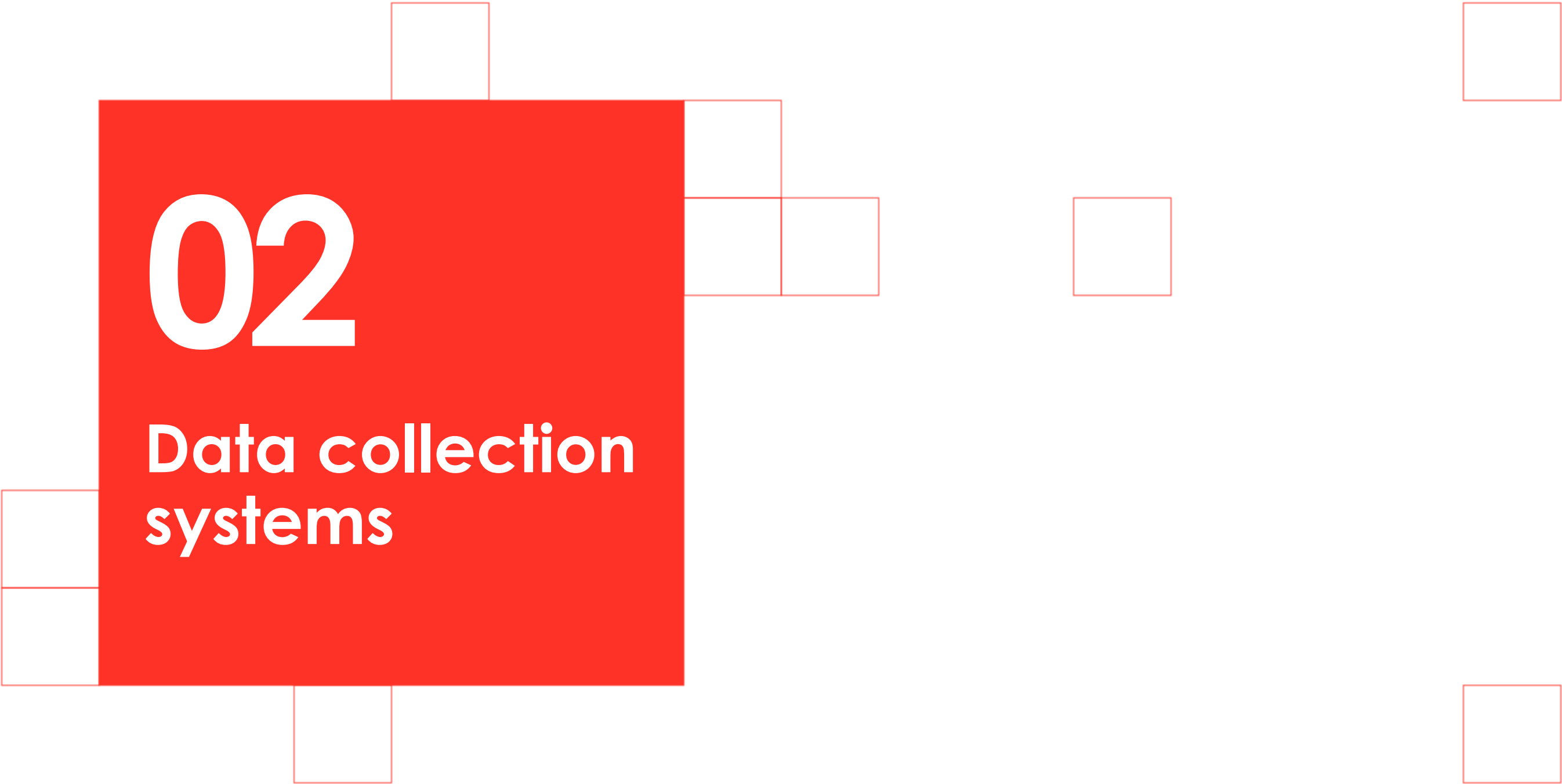
- **No planning capabilities** (can not handle long term manoeuvres or complex interactions with agents).
- Compounding errors: trained on expert data can lead to a snowfall of errors. An error of a poorly seen driving scene can lead to a cascade of errors that accumulate.
- Need for extensive and diverse amounts of data to tackle with Edge cases properly.
- Sensitivity of response to weather conditions: same driving scene can lead to different decisions.
- Adversarial perturbations and image poisoning: small changes on images can lead to huge changes on the decision taken from the network.

Naturalistic Driving

Some alternatives to PilotNet that can evolve .

Model	Core idea (1-liner)	Inputs	Output	Representative evidence / benchmark
CIL (2018)	Behavior cloning conditioned on high-level commands (left/right/straight) for intent controllability.	Mono camera + speed	Controls	ICRA'18; CARLA sim + 1/5-scale robot
CILRS (2019)	CIL with deeper ResNet and explicit speed branch; staple CARLA baseline.	Mono camera + speed	Controls	Common CARLA baseline in literature
ChauffeurNet (2019)	Mid-to-end trajectory policy with perturbations + environment losses to penalize collisions/off-road.	BEV map + detected actors + route	Waypoints / trajectory	ICLR/RSS'19; sim + real-car demo
LBC (2019)	Distill privileged planner (teacher) into a pure vision policy (student) for robust closed-loop driving.	Cameras (teacher uses GT)	Waypoints/controls	CoRL'19; 100% success on original CARLA benchmark
World on Rails (2021)	Model-based from logs: learn a forward world model; compute action values to supervise a policy.	Cameras + logs	Waypoints/controls	ICCV'21; #1 CARLA leaderboard at publication (40x less data)
TransFuser / ++ (2021-23)	Transformer fusion of multi-cam + LiDAR (PV $\leftrightarrow$ BEV attention) for robust closed-loop driving.	Multi-cam + LiDAR	Waypoints / trajectory	CVPR'21 / PAMI'23; top CARLA scores
InterFuser (2022)	Interpretable sensor-fusion transformer with safety controller constraining actions to safe sets.	Multi-cam + LiDAR	Waypoints + safety intermediates	CoRL'22; first on CARLA leaderboard at release
UniAD (2023)	Planning-oriented unified model (perception, prediction, planning) with shared queries.	Multi-cam (typ.)	Trajectory	CVPR'23; SOTA planning/prediction on nuScenes
VAD / VADv2 (2023-26)	Vectorized scene representation with instance-level constraints for safe, efficient planning.	Multi-cam	Trajectory	ICCV'23; strong closed-loop CARLA; high FPS
DriveTransformer (2025)	Unified Transformer with task-parallel queries, sparse sensor tokens, streaming temporal memory.	Multi-cam ($\pm$ LiDAR/map)	Trajectory	ICLR'25; SOTA on Bench2Drive & nuScenes reports



A decorative arrangement of squares in red and white. A large red square is the central element. To its left, two white squares are stacked vertically. Above the red square, one white square is centered. To the right of the red square, two white squares are stacked vertically, with another white square to the right of the top one. Below the red square, one white square is centered. Further to the right, one white square is centered. In the bottom right corner, there is a solid red square.

02

Data collection systems

Data Collection Systems

Training a Imitation Learning Neural Network typically requires gathering data from driver behaviour and from different sensors that populate a car.

Current commercial cars have massive ammount of sensors from ADAS systems.



Data Collection Systems

All data from ADAS sensors is sent to the CAN-bus for its consumption by the different computers that handle the vehicle.

The CAN-bus is a communication protocol for automóviles, where sensor data and commands from the different actuators (car computers, steering wheel, pedals...) travel.

All this information present at the CAN bus can be accessed in real time.

PROBLEM! Information travelling through the CAN bus is encrypted.



Data Collection Systems

OpenPilot, is an open source Project for autonomous driving intended for research, where a full Linux-based Operating System aimed for autonomous driving.

<https://github.com/commaai/openpilot>

This Project has a device (Comma-3X) that is that connects to the CAN-bus and reads messages from the can bus and allows for sending command messages to the CAN-bus allowing to take the control of the car.

Comma-3X has a lightweight Snapdragon processor (Same as Samsung S25) and wifi connection for data uploading and remote device handling.



Data Collection Systems

Car telemetry can be read as a sequence of JSON records.

Each JSON record is sampled at 30Hz

```

log: carControl
{
  'carControl': {
    'activeDEPRECATED': False,
    'actuators': {
      'accel': 0.05927348881959915,
      'brake': 0.0,
      'curvature': -0.0011517050443217158,
      'gas': 0.0,
      'longControlState': 'pid',
      'speed': 0.0,
      'steeringAngleDeg': 0.0,
      'torque': 0.07974967360496521,
      'torqueOutputCan': 0.0,
    },
    'angularVelocity': [-0.00041938901995308697, 0.0018860031850636005, -0.001347021316178143],
    'brakeDEPRECATED': 0.0,
    'cruiseControl': {
      'accelOverrideDEPRECATED': 0.0,
      'cancel': False,
      'override': False,
      'resume': False,
      'speedOverrideDEPRECATED': 0.0,
    },
    'currentCurvature': 20.0,
    'enabled': True,
    'gasDEPRECATED': 0.0,
    'hudControl': {
      'audibleAlert': 'none',
      'lanesVisible': True,
      'leadDistanceBars': 1,
      'leadVisible': False,
      'leftLaneDepart': False,
      'leftLaneVisible': True,
      'rightLaneDepart': False,
      'rightLaneVisible': True,
      'setSpeed': 5.5555534362793,
      'speedVisible': True,
      'visualAlert': 'none',
    },
    'latActive': True,
    'leftBlinker': False,
    'longActive': True,
    'orientationNED': [-0.0007602048572152853, 0.033073872327804565, 0.6949481964111328],
    'pitchDEPRECATED': 0.0,
    'rightBlinker': False,
    'rollDEPRECATED': 0.0,
    'steeringTorqueDEPRECATED': 0.0,
  },
  'logMonoTime': 117441725061,
}
  
```

log: carState

```

{
  'carState': {
    'aEgo': 0.1670699417591095,
    'accFaulted': False,
    'blockPcmEnable': True,
    'brake': 0.0,
    'brakeHoldActive': False,
    'brakeLightsDEPRECATED': False,
    'brakePressed': False,
    'buttonEnable': False,
    'buttonEvents': [],
    'canErrorCounter': 1,
    'canRcvTimeoutDEPRECATED': False,
    'canTimeout': False,
    'canValid': True,
    'carFaultedNonCritical': False,
    'charging': False,
    'clutchPressedDEPRECATED': False,
    'cruiseState': {
      'available': True,
      'enabled': True,
      'nonAdaptive': False,
      'speed': 0.0,
      'speedCluster': 0.0,
      'speedOffsetDEPRECATED': 0.0,
      'standstill': False,
    },
    'cumLagMs': 52.898983001708984,
    'doorOpen': False,
    'engineRpmDEPRECATED': 0.0,
    'espActive': False,
    'espDisabled': False,
    'eventsDEPRECATED': [],
    'fuelGauge': 0.0,
    'gasDEPRECATED': 0.0,
    'gasPressed': False,
    'gearShifter': 'drive',
    'genericToggle': False,
    'invalidLkasSetting': False,
    'leftBlindspot': False,
    'leftBlinker': False,
    'lowSpeedAlert': False,
    'parkingBrake': False,
    'regenBraking': False,
  },
}
  
```

Data Collection Systems

Front camera imaging is also captured



A decorative arrangement of squares with red outlines. A large solid red square is the central element. To its left are two stacked squares. Above it is one square. To its right are two stacked squares, with another square to the right of the middle one. Below it is one square. Further to the right is one square. In the top right corner is one square. In the bottom right corner is one square. In the bottom right corner of the slide is a solid red square.

03

Federated Learning

Federated Learning

Issues for training end-to-end naturalistic driving algorithms:

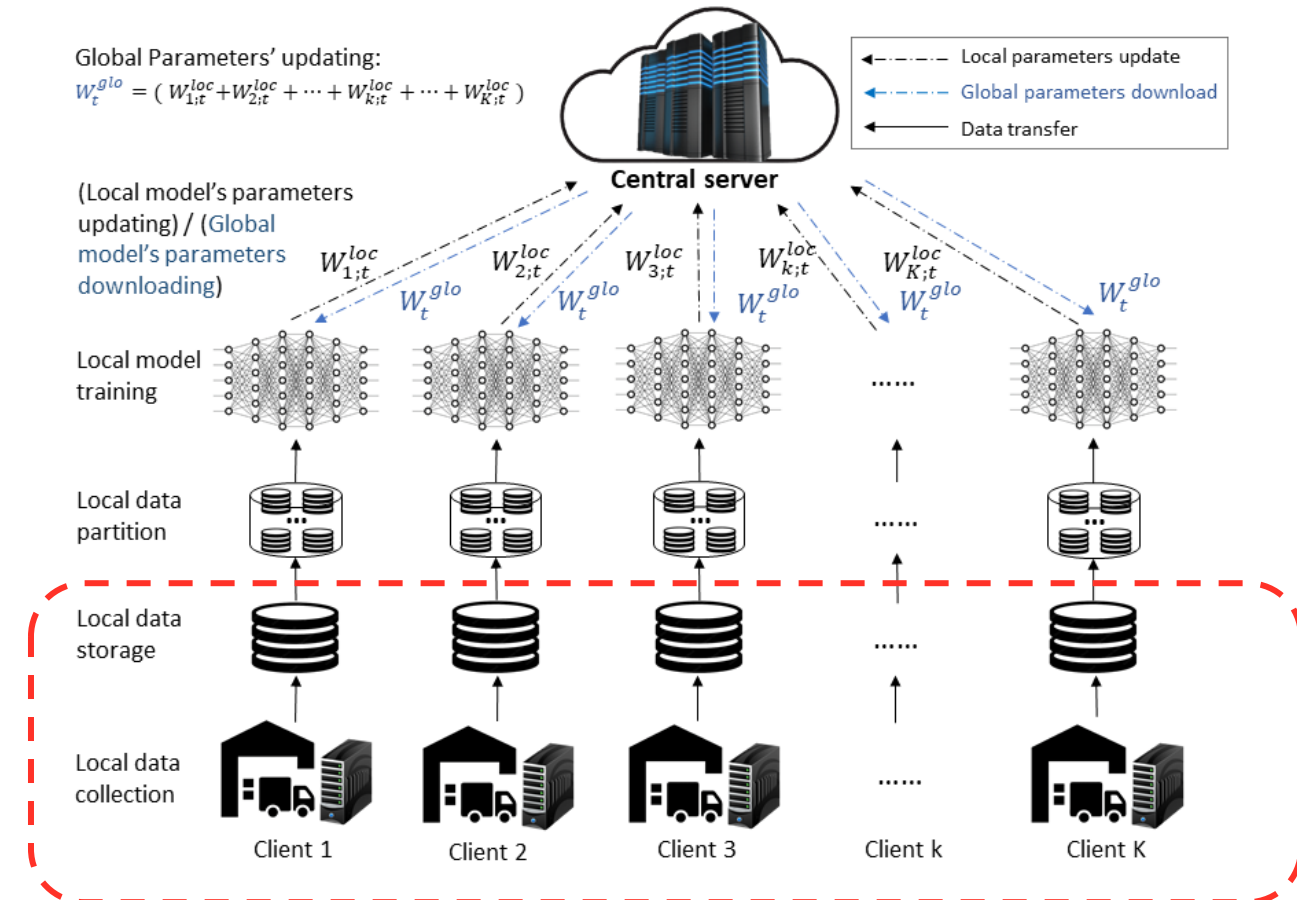
- **Privacy issue: can be inferred behavioral patterns from drivers. Habits can be unravel and housing and work location.**
- Latency issue: One hour of driving data can be several gigabytes, that can be hard to transmit to on mobile networks for connected cars.
- Lack of interoperability: Different brands/cars, may have different sensor suites or different CAN bus messaging style. This becomes terrible for models/brands with low units sold.

Federated Learning

Federated Learning is a part of Privacy Preserving Machine Learning (PPML).

At Federated Learning, data instead of being collected at a centralized datalake, data is spread across many data silos.

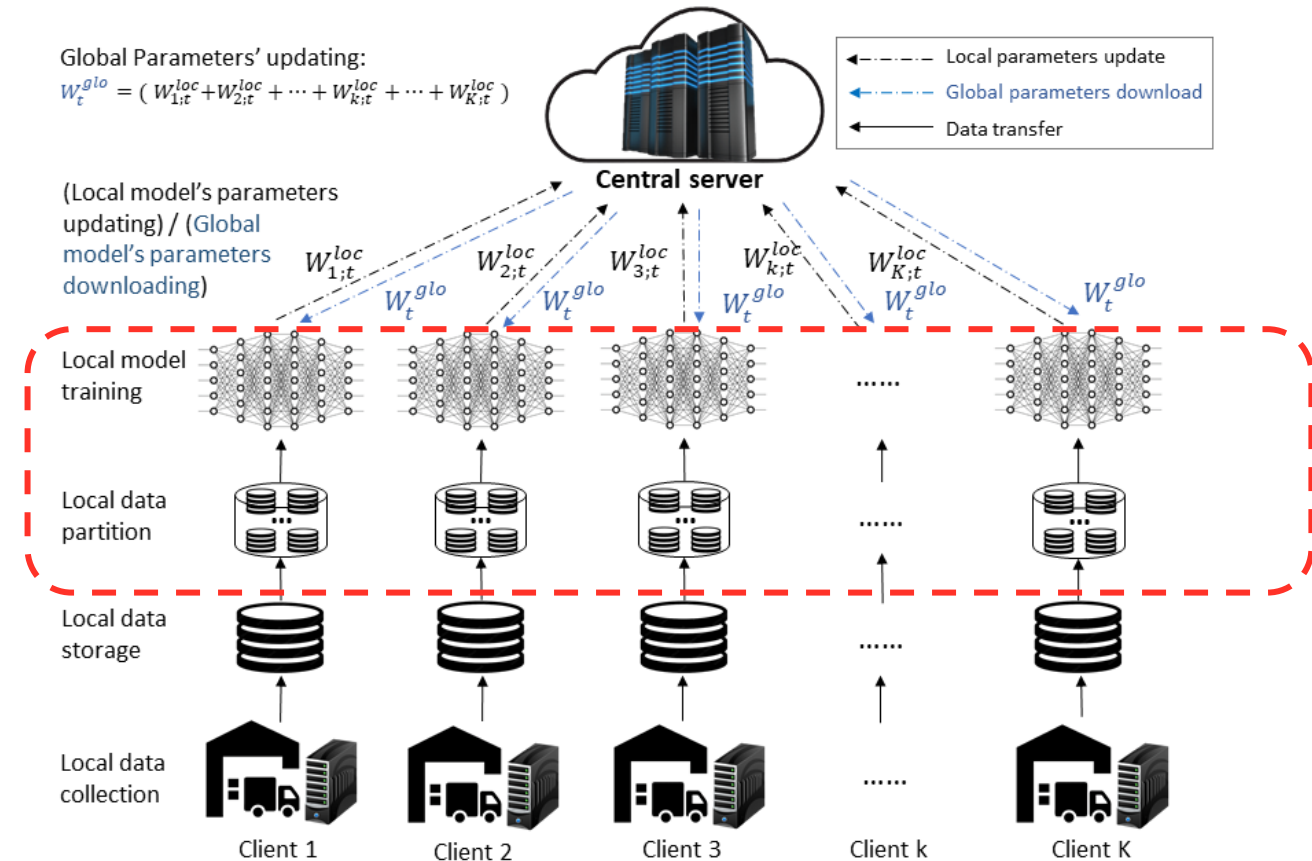
At Federated Learning data remains locked inside each of the data silos and never leaves them physically.



Federated Learning

Within each silo, there is a local partition of the data, where the Neural Networks are trained.

Thus, at each local partition an independent model is trained. Thus, there are as many weights for the Neural Network as local partitions.



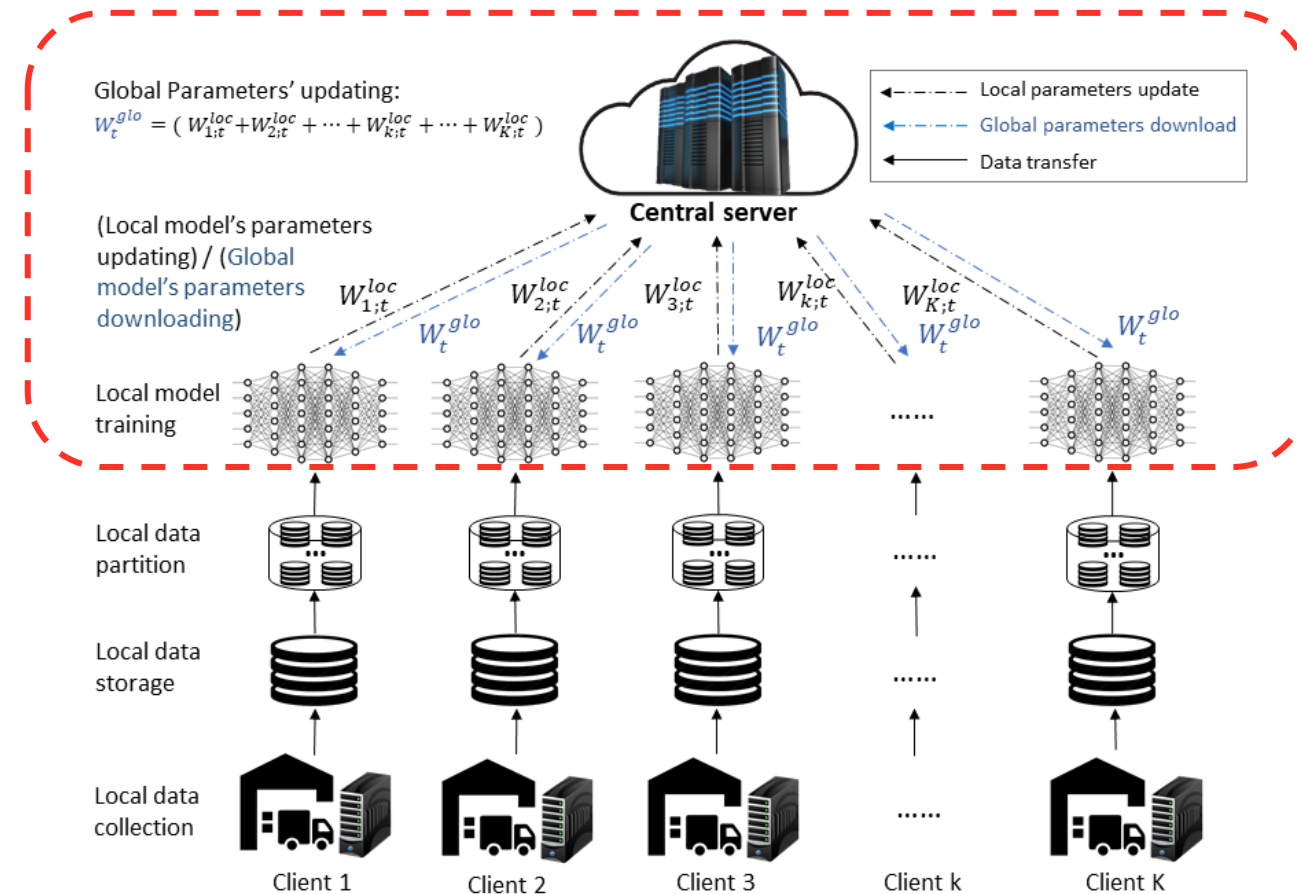
Federated Learning

The weights of each local Neural Network is then sent to a central server.

At the central server, all the weights of all the Neural Networks are combined into a single aggregated global model.

The aggregated global model combines the knowledge of all the data of all independent silos.

Federated Learning can combine data across different data silos without compromising data privacy and ownership.



Federated Learning

Typical aggregation functions for federation Will involve several ways of averaging weights.

Nevertheless, the simple mean (FedAvg) remains the most widely used.

Aggregation rule	Equation
FedAvg (weighted mean)	$w^{(t+1)} = \sum_{k=1}^K \frac{n_k}{N} w_k^{(t+1)}$
Coordinate-wise Median	$[w^{(t+1)}]_j = \text{median}([w_1^{(t+1)}]_j, \dots, [w_K^{(t+1)}]_j)$
Trimmed Mean	$[w^{(t+1)}]_j = \frac{1}{K-2f} \sum_{k=f+1}^{K-f} [w_{\pi_j(k)}^{(t+1)}]_j$

Notation: K clients; client k has n_k samples; $N = \sum_k n_k$; $w_k^{(t+1)}$ is client model after local training in round t ; π_j sorts the j -th coordinate values across clients; f is trimming level.



Federated Learning

Federated learning mitigates the previous issues by:

- Driving data never leaves the car, so data breaches are not possible.
- Weights of trained neural networks have a size of a few 100Mb. Thus, they can be transmitted across mobile networks.
- By exchanging only the weights, different types of sensors, the adaptation of the neural network remains a task for the car manufacturer.
- Averaging the weights allow to smear out outlier behavior of bad drivers.

Federated Learning enables a interoperable and modular framework for auto-manufacturers for an agnostic way to train in a collaborative manner Autonomous Driving algorithms.



Federated Learning

But Federated Learning may suffer from several issues that are still open problems at Autonomous Driving:

- Individually-trained algorithms are replaced by a global model. This model may not be suitable for all people by style of driving (i.e. too aggressive, too slow)
- Data silos are inherently multi-modal, need to study how to deal with data silos with low data:
 - Driver
 - Vehicle
 - Location (i.e. here is were Rovaniemi – Madrid make a Good federation).

Federated Learning

There are several Open Source Federated Learning Frameworks that can be used with Python.

Most relevant as of today are:

- Flower (<https://flower.ai/> <https://github.com/adap/flower>)



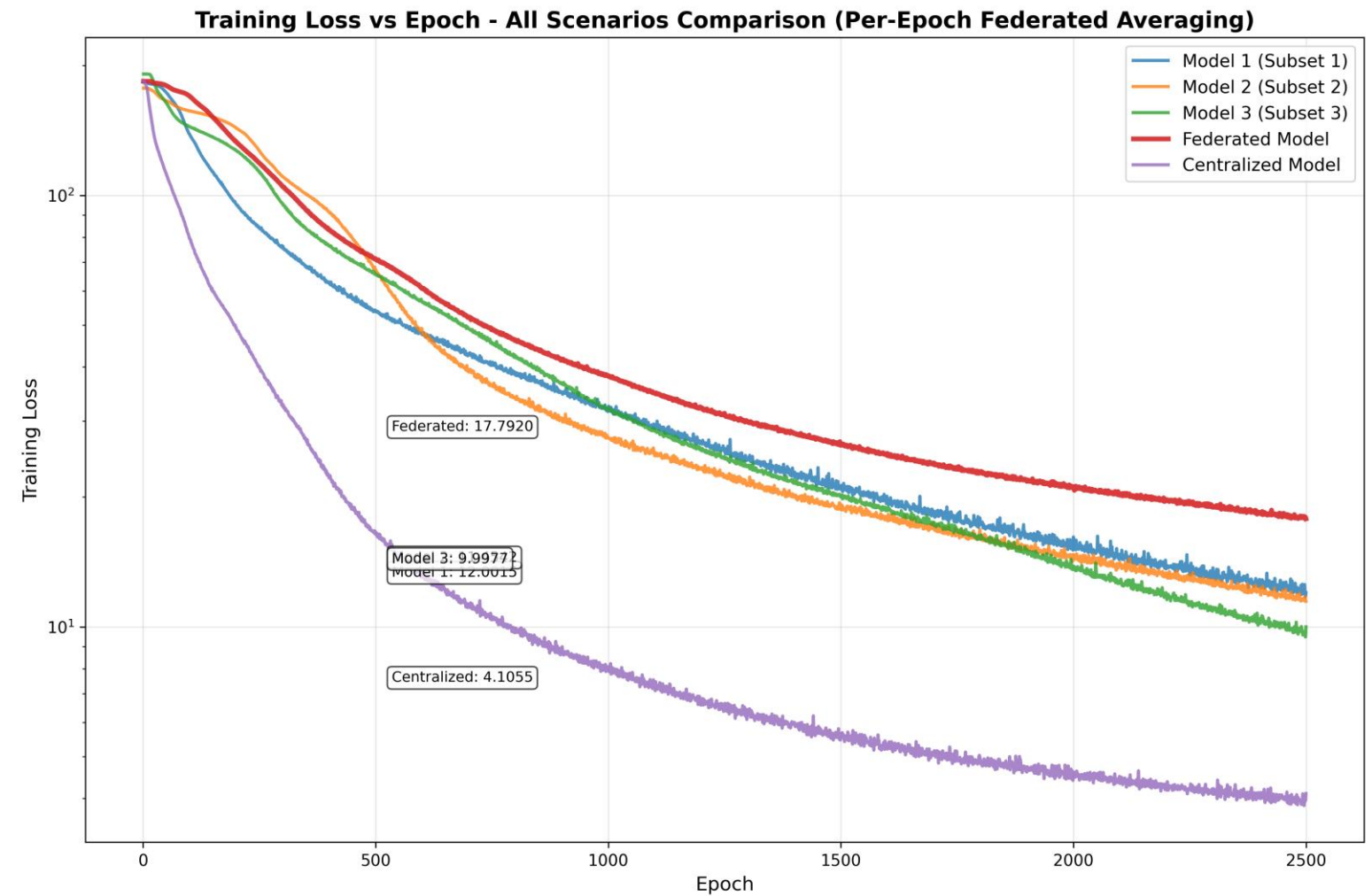
- PySyft (<https://openmined.org/pysyft/> <https://github.com/OpenMined/PySyft>).



Federated Learning

Some example of outcome from Federated Learning from Autonomous Driving in Naturalistic Driving scenarios.

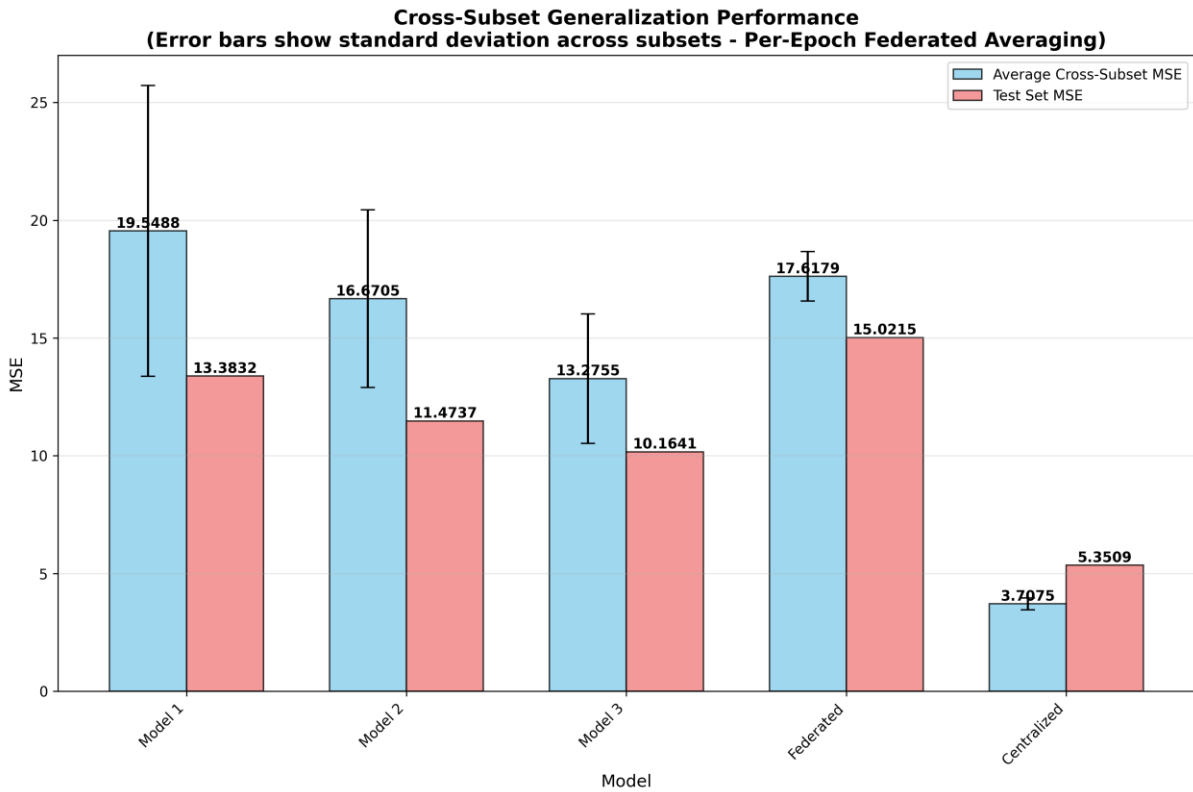
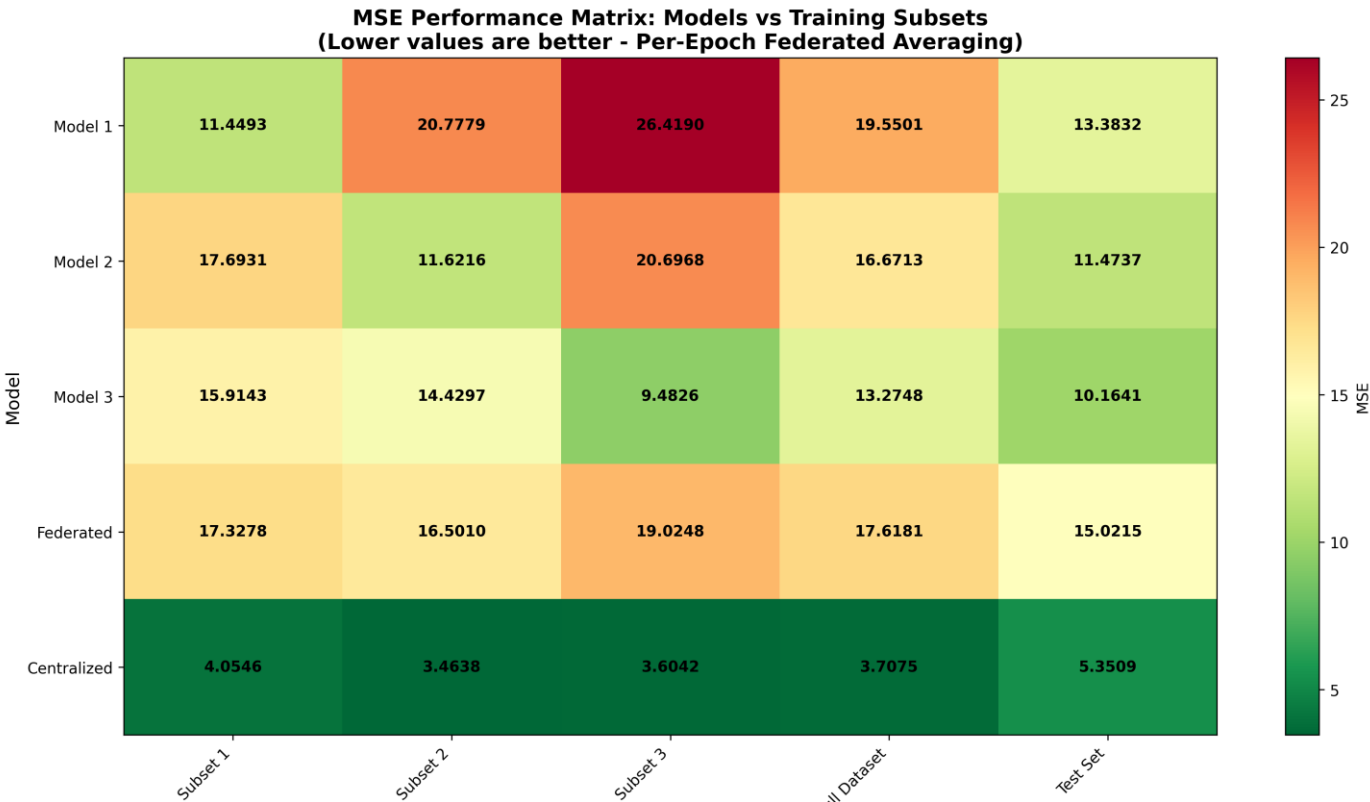
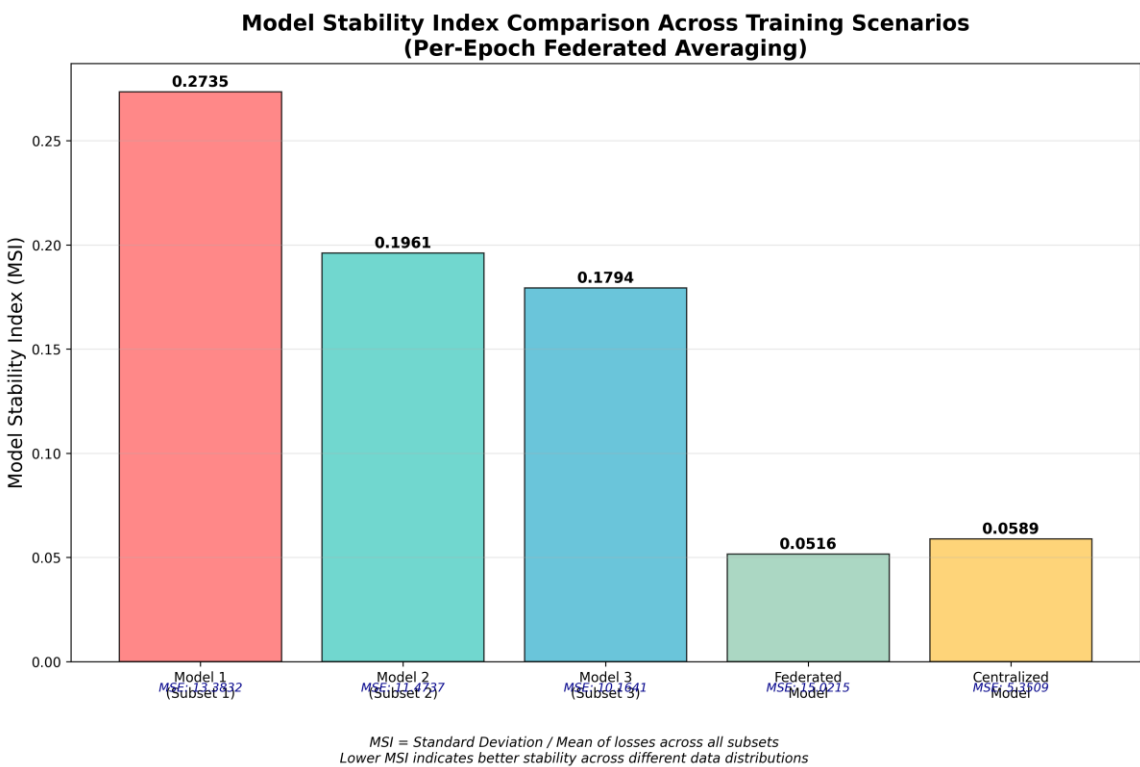
Toy example of a partition of 3 sub-sets a squence of 2h of driving.



Federated Learning

Although the centralized dataset remains unbeaten in terms of MSE, Federated models remains the one with the best stability across subsets.

Federated model is able to generalize better tan models trained on isolated silos.



Bonus: XAI

For critical systems, it is not only necessary to have a Good MSE. The reasoning of the model must be correct.

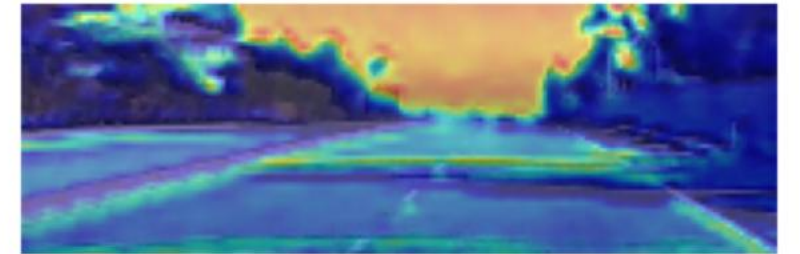
Due to regulatory issues and accountability in case of accident, an explanation of what elements of the driving scene has been taken into account for taking the decision must be made.

Techniques like Shapley coefficient and GradCAM can unravel the relevant elements of the scene.

Raw Image 1



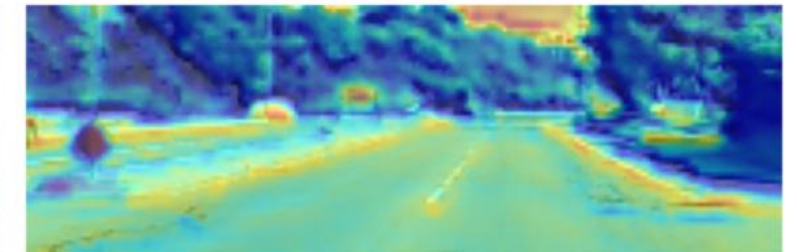
GradCAM 1



Raw Image 2



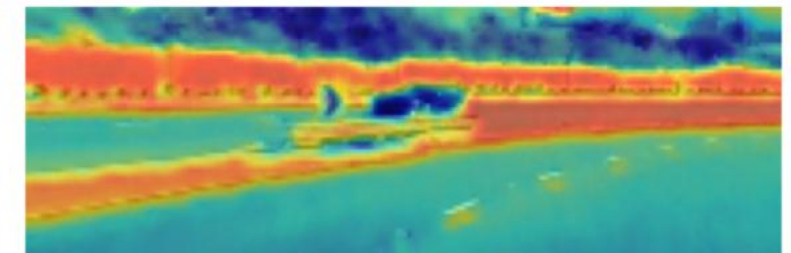
GradCAM 2



Raw Image 4



GradCAM 4





**Universidad
Europea**

Thanks

Dr. Manuel García Fernández

manuel.garcia2@universidadeuropea.es

Ve más allá